

HP IO Accelerator Driver and Management Software Versions 4.1.2 and 3.9.0 Release Notes

Abstract

This document describes details about the 4.1.2 HP IO Accelerator driver and 3.9.0 HP Accelerator Management Tool releases. This document also covers issues that might arise using these releases.



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Contents

About this guide	4
Description	4
System requirements	5
Hardware requirements	5
Sufficient system memory (RAM)	5
Supported operating systems	6
Supported Linux operating systems	6
Supported Windows operating systems	6
Supported VMware operating systems	7
Supported Solaris operating systems	7
Supported devices	7
IO Accelerator SX300 devices	7
IO Accelerator PX600 devices	8
Upgrade notes	9
Overview	9
Firmware version	9
Do not downgrade device firmware	9
Change log	10
4.1.2 IO Accelerator	10
Windows changes	11
Linux changes	11
Solaris changes	12
3.9.0 IO Accelerator Management Tool	12
Errata	14
4.1.2 IO Accelerator known issues	14
General issues	14
Management-specific issues	15
Windows-specific issues	15
Linux-specific issues	16
VMware-specific issues	19
Solaris-specific issues	20
3.9.0 IO Accelerator Management Tool known issues	20
Acronyms and abbreviations	23
Documentation feedback	24

About this guide

Description

This document describes the following information about the 4.1.2 IO Accelerator driver release:

- System requirements (on page 5), including supported operating systems and hardware requirements
- Supported devices (on page 7)
- Upgrade notes (on page 9), including the firmware version required for this release
- Change log (on page 10), including changes since the last generally available release
- Errata (on page 14), including issues that might arise using this release



CAUTION: Before upgrading to 3.x.x software and firmware, back up all data on the IO Accelerator. The 4.1.2 software and firmware reformat the drive, which causes data to be lost if not backed up. The 4.1.2 software is not backward compatible with 1.2.x or 2.x software.

This document also describes the following information about the 3.9.0 IO Accelerator Management Tool:

- Change log (on page 10), including changes since the last generally available release
- Errata (on page 14), including issues that might arise using this release

System requirements

Hardware requirements



IMPORTANT: For complete hardware requirements and installation instructions, see the *HP IO Accelerator Hardware Installation Guide*.

Sufficient system memory (RAM)

The amount of RAM the driver requires varies, according to the average block size written to the device. You can estimate the amount of system memory needed using the average block size in "Maximum RAM requirements (on page 5)."

Depending on your operating system, you can reduce worst-case memory use by formatting the IO Accelerator device with a 4096-byte sector size and force the average written block size to be 4096 bytes (4 K) or greater. However, some operating systems do not allow 4 K sector sizes.

HP IO Accelerator SX300 and PX600 devices ship with 4 K sector sizes.



CAUTION: Some operating systems only support 512 byte sector sizes. These operating systems include ESX and ESXi. For more information, see the "fio-format" section in the *HP IO Accelerator User Guide* for your operating system.



IMPORTANT: Microsoft does not support 4 K sector sizes with Windows Server 2008 R2; however, 4 K sector sizes do work with many applications. The performance benefit of 4 KB sectors is significant enough in Windows operating systems that testing 4 K sectors for use with your application is recommended. Microsoft does support 4 K sector sizes on Windows Server 2012.

Even if you cannot force 4 K sector sizes, the average write I/O size for most workloads is 4 KB or larger. For this reason, a 4 KB average write size is typically the most accurate representation of memory utilization.

Maximum RAM requirements

The amount of RAM required by the IO Accelerator VSL software depends on the IO Accelerator device capacity, formatted sector size, and how it is used.

The following table describes the upper limit of RAM that might be required of your system in a worst-case scenario.



IMPORTANT: The amount of RAM used by the IO Accelerator VSL software depends on your use case. The table below includes the worst-case numbers. Actual RAM usage will likely be less than the amount listed. You can run `fio-status -a` on the command line to see how much RAM the IO Accelerator VSL software is using per IO Accelerator device.

Device capacity	4096-byte blocks	2048-byte blocks	1024-byte blocks	512-byte blocks
1.0 TB	2.4 GB	4.7 GB	9.2 GB	18.2 GB

Device capacity	4096-byte blocks	2048-byte blocks	1024-byte blocks	512-byte blocks
1.25 TB	3.0 GB	5.8 GB	11.4 GB	22.6 GB
1.3 TB	3.1 GB	6.0 GB	11.8 GB	23.5 GB
1.6 TB	3.7 GB	7.3 GB	14.5 GB	28.8 GB
2.6 TB	6.0 GB	11.8 GB	23.5 GB	46.9 GB
3.2 TB	7.3 GB	14.5 GB	28.8 GB	57.8 GB
5.2 TB	11.7 GB	23.3 GB	46.4 GB	92.6 GB
6.4 TB	14.3 GB	28.5 GB	59.9 GB	113.7 GB

For example, if your system is equipped with a device that has a total capacity of 3,200 GB formatted to use 4,096 byte sectors, then your system might require as much as 7.3 GB of system RAM.

Supported operating systems

All operating systems must be 64-bit x86 architecture to support IO Accelerator devices. HP strongly recommends that you run the latest Service Pack of a release.

Supported Linux operating systems



IMPORTANT: Some distribution versions might have binary packages available for download. If your version does not have a binary package available, you can build the installation package from the available source package. Check the download folders for available packages. The following distributions are supported.

- Red Hat Enterprise Linux 5 (up to 5.11), 6 (up to 6.6), 7.0
- SUSE Linux Enterprise Server (SLES) 11, 11 SP2, 11 SP3, 12
- Oracle Linux 5 (up to 5.10), 6 (up to 6.5), 7.0
- CentOS 5 (up to 5.10), 6 (up to 6.5)
- Ubuntu 12.04, 14.04

Supported Windows operating systems

- Microsoft Windows 2008 Server 64-Bit, R2 (with SP1)
- Microsoft Windows 2012 Server
- Microsoft Windows 2012 Server R2
- Microsoft Hyper-V Server 2012
- Microsoft Hyper-V Server 2012 R2



IMPORTANT: IO Accelerator devices cannot be used as hibernation devices.

Hyper-V support

Hyper-V, as a Type 2 hypervisor on top of Windows Server 2008 R2, Windows Server 2012, or Windows Server 2012 R2 is supported.



IMPORTANT: With Hyper-V on Windows 2008 R2, only a 512 B sector size is supported on IO Accelerator devices. For more information on sector sizes on Windows operating systems, see the Microsoft Knowledge Base article (<http://support.microsoft.com/kb/2510009>).

Supported VMware operating systems

- VMware ESXi 5.0
- VMware ESXi 5.1
- VMware ESXi 5.5



IMPORTANT: All ESXi updates are supported unless otherwise specified.



IMPORTANT: Only SCSI versions of the IO Accelerator VSL software for ESXi are supported.

IO Accelerator devices are only compatible with operating systems that are 64-bit x86 architecture. This requirement means the following scenarios are supported:

- Using the IO Accelerator device as a VMFS datastore within the hypervisor and then sharing that storage with guest operating systems. Guest operating systems can be 32-bit or 64-bit because they are not directly using the IO Accelerator device.
- Using VMDirectPathIO to allow a VM to directly use the IO Accelerator device. In this case, only supported operating systems can use the device. VMDirectPathIO is currently supported on Windows and Linux operating systems that are supported on HP IO Accelerator devices.

The VSL software is only required for ESX or ESXi if you plan to use the IO Accelerator device as a VMFS datastore. If you are passing the devices through using VMDirectPathIO, you do not need to install the VSL on your the ESX or ESXi system. Instead, install the VSL on the guest system. For example, pass the device through to a Windows VM, and then install the Windows VSL on that VM.

For installation instructions, see the *HP IO Accelerator User Guide*.

Supported Solaris operating systems

- Solaris (x86, 64-bit) 10 Update 11
- Solaris (x86, 64-bit) 11, 11.1

The IO Accelerator VSL software does not support SPARC processors.

Supported devices

This section lists the HP devices that are supported with this version of the IO Accelerator driver.

IO Accelerator SX300 devices

- SX300-1300
- SX300-1305
- SX300-1600

- SX300-1605
- SX300-3200
- SX300-3205
- SX300-6400
- SX300-6405
- SX300 Type B Mezzanine Cards

IO Accelerator PX600 devices

- PX600-1000
- PX600-1005
- PX600-1300
- PX600-1305
- PX600-2600
- PX600-2605
- PX600-5200
- PX600-5205

Upgrade notes

Overview

This version of the IO Accelerator VSL software only supports IO Accelerator PX600 devices and SX300 devices. The IO Accelerator VSL software does not support devices that are compatible with IO Accelerator VSL software version 3.x.x or earlier (for example, ioDrive2, ioDrive, ioScale2, or ioScale devices are not supported).

Firmware version

This version of the IO Accelerator driver is compatible with firmware archive version 4.1.2-20141212.

HP recommends that you back up any data on the device before performing the upgrade. The latest supported version of the firmware and VSL can be found on the HP website (<http://www.hp.com>).

Do not downgrade device firmware

New IO Accelerator devices are shipped with recent firmware versions, often the minimum version required for the device.



CAUTION: Do not downgrade the IO Accelerator device to an earlier version of the firmware. Earlier versions of the firmware might not be compatible with the device, and downgrading the firmware might result in data loss. If you must downgrade your firmware, contact HP Customer Support (<http://www.hp.com/support>) for compatibility information and to discuss your use case.

If you are installing new IO Accelerator devices in a system using older devices and firmware, upgrade the older devices to the latest firmware and driver as a best practice.

Change log

4.1.2 IO Accelerator

This change log documents the changes made from 4.1.1 to 4.1.2, which includes the following:

- Updated supported operating system. For more information, see "Supported operating systems (on page 6)."
 - Newly supported Linux operating systems:
 - CentOS 7.0
 - Oracle Linux 7.0
 - RHEL 6.6, 5.11
- Improved how multiple IO Accelerator VSL software utilities simultaneously access IO Accelerator devices.
- Augmented the fio-status utility JSON output to include FPGA firmware version.

Fixed the following issues:

- **Misleading debug messages**

Issue: The IO Accelerator VSL software printed misleading debug messages in the system log. For example:

```
Suspect pad <number> on bank <number> checked: at least <percentage> retired.
```

Resolution: The misleading messages no longer appear.
- **Rare powercut issue**

Issue: In the rare instance that the IO Accelerator VSL software erases a block on the device during a powercut event, the device might fail to properly write pending data. The device would then require a rescan or not properly attach.

Resolution: Devices now abort erases and properly handles the powercut flush.
- **Erroneous power throttling messages**

Issue: The IO Accelerator VSL software reported power throttling messages for IO Accelerator SX300 devices when the devices were not throttled.

Resolution: The erroneous messages no longer appear.
- **Issues with non-0 PCI domains**

Issue: The IO Accelerator VSL software utilities, especially the fio-pci-check utility, did not properly handle non-0 PCI domains.

Resolution: The utilities now handle multiple PCI domains.
- **Erroneous hardware configuration messages**

Issue: Erroneous error message "unable to configure hardware" with no actual hardware failure.

Resolution: Suppressed erroneous error message.
- **Ignored interrupts**

Issue: Due to a race condition, the IO Accelerator VSL software could ignore a hardware interrupt for up to 10 ms.

Resolution: The race condition is removed.

- **Channel failures causing crashes**

Issue: An operating system would crash when an IO Accelerator device experienced a channel failure in a system with multiple IO Accelerator devices.

Resolution: The kernel no longer crashes when an IO Accelerator device experiences this failure.

- **Kernel panics in minimal mode**

Issue: An unlikely kernel panic might occur when IO Accelerator devices are in minimal mode.

Resolution: The kernel no longer panics in this scenario.

Windows changes

Fixed Windows issues:

- **Irrelevant messages**

Issue: Irrelevant Windows driver memory management messages are displaying. For example:

```
,FusionEventDriver,4,(2),"Alloc memory chunk FFFFE022B600000, size = 134217728 bytes, pages = 32768"
```

Resolution: The messages no longer appear.

- **Multiple instances of fio-detach caused hangs**

Issue: Running the fio-detach utility on multiple devices at the same time in Windows would cause the utility to hang.

Resolution: You can now run multiple instances of this utility in parallel.

Linux changes

Linux changes:

Improved logging for invalid Linux I/O requests.

Fixed Linux issues:

- **Driver load issues**

Issue: Updating the system kernel or IO Accelerator VSL software would insert the IO Accelerator VSL software into the `initrd` and ignore the software configuration file (`/etc/modprobe.d/iomemory-vsl.conf`). This issue created software loading issues that required `dracut` to fix.

Resolution: Choose one of the following solutions to this configuration issue.

- If DKMS is installed and configured, the IO Accelerator VSL software is configured for the new kernel.
- If `dracut` is installed, the IO Accelerator VSL software uses that program to fix the configuration issue.
- If both the `fio-sysvinit` and `fio-common` packages are installed, the IO Accelerator VSL software is excluded from the `initrd`.

- **Watchdog poll counter**

Issue: The `saved_by_watchdog_poll` counter did not increment.

Resolution: The `saved_by_watchdog_poll` now increments properly.

- **Unnamed threads**

Issue: IO Accelerator VSL block threads were not matched to specific devices and the device names were replaced with question marks. For example, `fio?`:

```
root 25205 0.0 0.0 0 0 ? S Aug28 0:00 [fio?-scan]
root 25206 0.0 0.0 0 0 ? S Aug28 0:00 [fio?-sselect]
```

Resolution: The threads are now properly named.

- **Improper shutdowns**

Issue: Certain IO Accelerator device failures in combination with the parameter `use_work_queue=3` causes the IO Accelerator VSL software to fail to shut down properly.

Resolution: The IO Accelerator VSL software now shuts down properly under these conditions.

- **3.16 kernel support**

Issue: The IO Accelerator VSL software would not build for the 3.16 Linux kernel.

Resolution: The software will now build on the 3.16 kernel.

- **Build error**

Issue: Attempting to build the IO Accelerator VSL software for the `2.6.39- 400.215.10.el5uek` kernel or newer would fail with the following error:

```
WARNING: /lib/modules/2.6.39-
400.215.10.el5uek/extra/fio/iomemory-vsl4.ko needs unknown symbol
kfio_bio_alloc_chain
```

Resolution: The software now builds with this kernel and newer.

Solaris changes

Fixed Solaris issues:

- **Fixed CPU usage**

Issue: The reported IO Accelerator VSL software CPU consumption on Solaris was 100%.

Resolution: The reported/actual CPU consumption on Solaris is now decreased.

- **Issue when compiling the software**

Issue: The IO Accelerator VSL software for Solaris would fail to load due to an installation script error.

Resolution: The IO Accelerator VSL software now loads properly.

3.9.0 IO Accelerator Management Tool

This change log documents the changes made from 3.7.0 to 3.9.0, which includes the following:

Issue	Windows	Linux	ESXi
Changes have been made to allow Caching Management using IO Accelerator PX600 and SX300 devices.	X	X	X
IO Accelerator Management tool includes new licensing for use with caching software, including evaluation license handling.	X	X	X

Issue	Windows	Linux	ESXi
Issue with Traps received by SIM 7.3 showing as unregistered traps is resolved.	X	X	X
Discrepancies over UI showing that VMs are not configured for caching or caching status is resolved.	X	X	X
An issue resulting in a large number of "ERROR cluster_hnd_gen - Failed to clear cache for handle" errors is resolved.	X	X	X
SNMP polling was adversely affecting CPU usage when a large number of devices were used.	X	X	X
Enhancements have been made to provide more detail on <code>uber_monitor</code> error messages.	X	X	X
An issue was resolved to display the proper capacity utilization values when using custom values.	X	X	X
Improvements were made on the default cache grid.	X	X	X
The issue that occurred when closing the <code>fio-agent</code> caused session limits to be reached, rejecting <code>fio-agent</code> connections is resolved.	X	X	X
The issue of Linux VMs having guest-based caching enabled showing a status of "No Caching" is resolved.	—	X	—
The issue of low-level formatting requiring to be done within the vSphere client is resolved.	X	X	—
Live Graphs will now work with the installation of the host software.	X	X	—

Errata

4.1.2 IO Accelerator known issues

This section describes issues you might encounter when using the 4.1.2 IO Accelerator driver release.

General issues

Do not disable CPUs after loading the IO Accelerator VSL driver

If you plan to take any CPUs offline (including disabling Hyper-Threading Technology), you should do so before the IO Accelerator VSL driver loads and begins to use the available CPUs. If you disable any CPUs that were being used by the IO Accelerator VSL software, then the software might hang.

Keep default MSI-X for better performance

All IO Accelerator devices use MSI-X message signaled interrupts. This improves performance while decreasing CPU load. However, there are some compatibility issues with MSI-X in certain environments.

If you are experiencing performance issues, you might want to disable MSI-X. Doing so will then enable MSI interrupts by default. If you wish to use legacy interrupts instead, you may also disable MSI interrupts.

To disable either MSI-X or MSI, set the appropriate module parameter (`disable_msix` or `disable_msi`) value to 1. For more information, see the *HP IO Accelerator User Guide*.



IMPORTANT: To disable MSI in Windows, edit the MSI Supported registry entry. You cannot disable only MSI-X in Windows. If you need to disable MSI-X, you must disable both MSI and MSI-X by disabling MSI. For more information, see *Enabling Message-Signaled Interrupts in the Registry* on the Microsoft website ([http://msdn.microsoft.com/en-us/library/windows/hardware/ff544246\(v=vs.85\).aspx](http://msdn.microsoft.com/en-us/library/windows/hardware/ff544246(v=vs.85).aspx)).

Proper time on startup

If the IO Accelerator device does not boot up with the proper time set on the system, then the software might delay starting, because the driver self-tunes to the difference between the reflected age data and actual age of data.

If the time is set backward on a running system, this setting might result in decreased card performance for the lesser of one day or the amount the time is set backward.

Proper time is within a few minutes of actual time.

Management-specific issues

Be sure the utilities match the IO Accelerator VSL software version

When you install this version of the IO Accelerator VSL software, be sure that you install the utilities that go with this version. Each set of utilities is designed to work with a specific version of the IO Accelerator VSL software.

If you use a set of utilities that does not match the IO Accelerator VSL software, you might see an error in the command line or logs such as `unhandled ioctl` or `Error: This version of <utility> is not compatible with the running driver`. To solve this issue, reinstall the utilities using the package with the correct version number.

Utility failed while running `fio-bugreport`

The `fio-bugreport` utility uses other utilities to create the report. Depending on the operating system, some of these additional utilities might not be available and the `fio-bugreport` displays an error that a `fio` utility failed or was not found.

The `fio-bugreport` utility is designed to continue even if a component fails. The report is still created.

`fio-status` might not display failed devices

When an IO Accelerator device fails, occasionally, it does not appear in the `fio-status` report. If your device has failed, contact HP support (<http://www.hp.com/go/assistance>).

Windows-specific issues

SCSI ID conflict with other storage devices

If your Windows system uses a storage device in addition to an IO Accelerator device, the devices might have duplicate SCSI IDs that will cause conflicts. You can resolve the conflict by changing the IO Accelerator device SCSI ID using the `WIN_SCSI_BUS_ID` module parameter.

This parameter sets the Windows SCSI ID number for all IO Accelerator devices in the system to avoid conflicts with other SCSI device IDs. The default value, 0, is off and no IDs are set. Any value between 1–254 will set the SCSI IDs for all IO Accelerator devices in the system to that number. IO Accelerator devices do not use SCSI IDs directly, so any non-conflicting number will do.

For more information, see the *HP IO Accelerator User Guide*.

Specific partitions required for devices with capacities greater than 2 TB

Devices with capacities greater than 2 TB, such as the 3.0 TB MLC ioDrive2 device, require the following partition types:

- Single device: GPT (GUID Partition Table)
- Multiple devices (for a RAID configuration): Dynamic Disk

These devices also require sector sizes greater than 512 B (HP recommends 4 KB sectors). When you format these devices using `fio-format`, the default sector size is 4 KB.

IO Accelerator VSL software not loading or attaching devices after install

If the IO Accelerator VSL software is not loading or attaching IO Accelerator devices after installation (including an upgrade), make sure that you have rebooted the system after the installation.

If a reboot does not solve the problem, follow the manual installation procedure in the appendix of the user guide for Windows. Repeat this procedure to install each device.

Conversion to GPT or Dynamic disk terminates Logical Disk Manager Admin Service

This issue appears in any of these cases:

- Converting a Basic partition to GPT
- Converting a Basic partition to Dynamic Volume
- Switching between GPT and Dynamic Volume, either way

The message is: The Logical Disk Manager Administrative Service terminated unexpectedly. Restart the 'Virtual Disk' service. This problem occurs only with Windows Server 2003.

If the IO Accelerator devices are to be used in GPT or Dynamic mode, the following process should be done during the initial setup. This process also recovers devices that have had a failed conversion attempt.



CAUTION: To avoid loss of data, be sure to back up any existing data on your devices. This procedure destroys any existing data on your devices.

1. For each IO Accelerator device in the system that is to be converted, use the management software to detach the drive and perform a low-level format. For more information, see the *HP IO Accelerator Management Tool User Guide*.
2. Restart the computer.
3. Go to Disk Management and then select **Initialize Disk**.
4. Right-click and then select **Convert to GPT** (or **Convert to Dynamic Disk**).

Linux-specific issues

dracut: not found error during installation

When you install the IO Accelerator VSL software on a system that does not have `dracut` installed, you will see the following error:

```
# rpm -ivh iomemory-vsl4-<version>.rpm
Preparing... ##### [100%]
1:iomemory-vsl4-<version>-39#####
[100%]
/var/tmp/rpm-tmp.52200: line 12: hash: dracut: not found
```

You can safely ignore this message. The IO Accelerator VSL software does look for the `dracut` utility to assist in loading the IO Accelerator VSL software, but the utility is not required. For more information on how software loads, see the Linux changes (on page 11) in the Change log (on page 10).

Error messages when installing IO Accelerator VSL software on RHEL 7.0

When the IO Accelerator VSL software is installed on RHEL 7.0, you might see the following errors:


```
dracut-install: ERROR: installing 'vi'
dracut-install: ERROR: installing '/etc/virc'
/usr/lib/dracut/dracut-install -D /var/tmp/initramfs.pcKkj9 -a vi /etc/virc
ps
grep cat rm
rm: cannot remove '/var/tmp/dracut-log.LctCNA': Directory not empty
```

This is a known issue with RHEL 7.0, and, despite the errors, the software should install correctly. For more information on this issue, see the bug report on the Red Hat website (https://bugzilla.redhat.com/show_bug.cgi?id=1118988).

The 3.7 kernel is incompatible with the IO Accelerator VSL software

The 3.7 Linux kernel has a known issue that makes it incompatible with the IO Accelerator VSL software. You can fix the kernel by applying the patch available on the Patchwork website (<https://patchwork.kernel.org/patch/2031631/>).

fio-firmware might incorrectly report pending firmware

In the rare case when firmware version information gets corrupted, fio-firmware will start reporting that there is pending firmware, even when there isn't. Use fio-status to confirm there really is pending firmware.

Upgrading the Kernel in Linux

To upgrade the kernel when the IO Accelerator VSL software is installed:

1. Unload the IO Accelerator VSL driver.
2. Uninstall the IO Accelerator VSL software.
3. Upgrade the kernel.
4. Install an IO Accelerator VSL software package that is compiled for the new kernel.

Failure to follow this procedure might result in driver load issues.

Compiler Cache (ccache) causes IO Accelerator VSL software src.rpm rebuild failures on some distributions

If the ccache package is installed, rebuilding the IO Accelerator VSL software src.rpm might fail with an error similar to the following:

```
CC [M]
/root/fio/iomemory-vls-<version>/root/usr/src/iomemory-vsl/driver_init.o
/root/fio/iomemory-vls-<version>/root/usr/src/iomemory-vsl/driver_init.c
:116:
error: initializer element is not constant
[...]
```

To allow the VSL to rebuild, remove the ccache package or disable ccache.

Rare error on driver unload using kernels older than 2.6.24

A bug in Linux kernels prior to 2.6.24 can cause a general protection fault or other kernel error when the IO Accelerator driver is unloaded. This bug also affects non-HP drivers. The bug has been resolved in newer kernels.

Because this is a bug in the Linux kernel, HP cannot resolve this issue for older kernels.

ext4 in Kernel.org 2.6.33 or earlier might silently corrupt data when discard (trim) is enabled



CAUTION: HP does not support the use of ext4 in Kernel.org 2.6.33 or earlier. Ext4 in Kernel.org 2.6.33 or earlier might silently corrupt data when discard is enabled.

The ext4 filesystem in the Kernel.org kernel 2.6.33 and earlier contains a bug where the data in a portion of a file might be improperly discarded (set to all 0x00) under some workloads. Use Version 2.6.34 or newer to avoid this issue. For more information, see the patch

(<http://git.kernel.org/?p=linux/kernel/git/torvalds/linux-2.6.git;a=commitdiff;h=b90f687018e6d6>) and bug report (https://bugzilla.kernel.org/show_bug.cgi?id=15579).

The fix is included in RHEL6 pre-release kernel kernel-2.6.32-23.el6. The eventual release RHEL6 kernel is not affected by this issue.

Discard support was added to the kernel.org mainline ext4 in Version 2.6.28 and was enabled by default. For fear of damaging some devices, discard was set to default to disabled in Version 2.6.33-rc1 and was back ported to 2.6.31.8 and 2.6.32.1. For more information, see the kernel patch

(<http://git.kernel.org/?p=linux/kernel/git/torvalds/linux-2.6.git;a=commitdiff;h=5328e635315734d>).

Kernels 2.6.34/35 don't handle switching interrupt types

Linux kernels around 2.6.34/35 might have problems processing interrupts if the driver is loaded using one interrupt type, unloaded, and then loaded again using a different interrupt type. The primary symptom is that the IO Accelerator device is unusable and the kernel logs have errors containing `doIRQ`. For example, the following sequence on an affected system would likely result in errors:

1. Load the driver with a default of `disable_msi=1` which selects APIC interrupts:

```
$ modprobe iomemory-vsl
$ modprobe -r iomemory-vsl
```

2. Load the driver and enable MSI interrupts:

```
$ modprobe iomemory-vsl disable_msi=0
```

To work around this issue, if you see the error, reboot the system. Also, always load with the same interrupt type selected. To change between interrupt types, reboot the system first.

RHEL6 udevd warning

When using an IO Accelerator under RHEL6 (or any Linux distribution with `udev` script version 147 or later), `udev`d might emit the following messages:

```
udev[154]: worker [19174] unexpectedly returned with status 0x0100
udev[154]: worker [19174] failed while handling
'/devices/virtual/block/fioa'
```

These messages are innocuous, and you can ignore them.

RHEL6 gives a `warn_slowpath` message during device attach

When attaching an IO Accelerator device under RHEL6, you might see log messages similar to the following:

```

kernel: -----[ cut here ]-----
kernel: WARNING: at fs/fs-writeback.c:967 __mark_inode_dirty+0x108/0x160()
(Tainted: P ----- )
.
.
.
[<ffffffff8106b857>] warn_slowpath_common+0x87/0xc0
[<ffffffff8106b8aa>] warn_slowpath_null+0x1a/0x20
.
.
.

```

This is due to a bug in the 2.6.32 kernel, and the messages can safely be ignored.

Switching interrupt types with newer kernels might cause errors

With newer Linux kernels, switching interrupt types after initial driver load might cause the kernel to report `doIRQ` errors. As a workaround, reboot your system before loading the driver with the new interrupt type specified.

Do not use an IO Accelerator device as a kdump target

Do not direct `kdump` to dump the crash information to an IO Accelerator device. The IO Accelerator VSL software does not load in the `kdump` crashkernel, and IO Accelerator devices are not supported as `kdump` targets.

VMware-specific issues

ESXi 5.5 rollup driver installer

The Rollup Driver installer for ESXi 5.5 includes version 3.2.6 of the IO Accelerator VSL software. If you use this version of the ESXi 5.5 installer, you need to uninstall IO Accelerator VSL version 3.2.6 before you install any newer version. For more information, see the *HP IO Accelerator User Guide*.

Only 512 B sectors supported

VMware hypervisors only supports a 512 B sector size. After you install the IO Accelerator VSL software, you must reformat the sectors to a 512 B size on IO Accelerator SX300 and PX600 devices before using the devices. For more information, see the `fio-format` section of the *HP IO Accelerator User Guide*.

The issue is resolved with the SCSI device version of the IO Accelerator VSL software. This issue is still seen in the raw block device version of the software.

ESXi 5.x injected installer allows installation on an IO Accelerator device

Do not install the host OS on an IO Accelerator device. IO Accelerator devices are not designed to be bootable. The ESXi injected installer permits you to install the OS on an IO Accelerator device, but the installation will fail on reboot.

IO Accelerator devices are only labeled as a VMware Block Device in the installer screen. Make sure to select a bootable device during the installation.

vCenter cannot manage extents on IO Accelerator devices

You cannot use vSphere vCenter to manage extents on IO Accelerator devices, including growing or spanning extents.

However, you can connect directly to the host using the vSphere client and manage extents on IO Accelerator devices.

Solaris-specific issues

Only 512 B sectors supported

Solaris only supports a 512 B sector size. After you install the IO Accelerator VSL software, you must reformat the sectors to a 512 B size on IO Accelerator SX300 devices prior to using the devices. For more information, see "fio-format" in the *HP IO Accelerator User Guide*.

No quiesce DDK interface

The IO Accelerator VSL software does not implement the quiesce DDK interface on Solaris. This means that the IO Accelerator device might continue to perform DMA operations after the IO Accelerator VSL driver is instructed to shutdown, which can result in data corruption if the memory has been reused as well as kernel panics if the IOMMU is enabled. To reduce the chance of data corruption, disable "fastboot" to give the IO Accelerator VSL driver a better chance to shutdown cleanly before reboot.

Solaris cannot use IO Accelerator devices with capacities greater than 2 TB

You cannot use the full capacity of some devices because Solaris 10 does not support devices with capacities greater than 2 TB. You need to down-format these devices to 2 TB using `fio-format`. For example,

```
fio-format -s 2T /dev/fctl.
```

Provide adequate RAM

The IO Accelerator VSL might use a large amount of system RAM. If enough RAM is not available, the system will crash. For RAM requirements, see "Hardware requirements (on page 5)."

As the system runs out of RAM, it might generate the following message:

```
verify: bad magic header 0, wanted acca at file /dev/rdisk/c6d0p0 offset
212606976, length 0
```

3.9.0 IO Accelerator Management Tool known issues

This section describes issues you might encounter using the IO Accelerator Management Tool 3.9.0 release.

Issue	Windows	Linux	ESXi
IO Accelerator Management Tool 3.9.0 is not backward compatible with ioTurbine 2.1.X.	X	X	X

Issue	Windows	Linux	ESXi
Microsoft driver certifications are no longer provided for Windows Server 2008. IO Accelerator Management Tool still runs correctly on Windows Server 2008, but Microsoft certificates will not be installed on the machine.	X	—	—
The latest version of IO Accelerator Management Tool might not correctly display the latest firmware in the Update firmware dialog box. If this occurs, use the Command Line Interface as detailed on page 16 of the ioSphere Tools Guide to upgrade the firmware.	X	X	X
With the upgrade in live graph functionality, some live graph renderings will appear to be erratic. The bar graphs displaying real-time data will jump around rapidly as the system attempts to render the data in real time. Where we used to display these bar graphs as running averages that settled quickly down to a more or less constant value, now the constantly changing nature of the incoming data causes the bar graph to appear more dynamic in nature.	X	X	X
Changing a VM's network adapter can cause VMs configured for guest caching to be displayed as not configured for caching. To fix this issue, restart the vme2 service on the management server.	X	X	X
Attempting to run IO Accelerator Management Tool in multiple browsers on the same host might cause unstable behavior in the tool.	X	X	X
In vSphere client and in Internet Explorer (configured for IE7 compatibility mode), clicking/moving the slider bar several times within the "Write Performance vs. Capacity" widget in the reformat dialog will cause the widget to disappear. Close the dialog and reopen it when this happens to reset the view.	X	X	X
With a very large vCenter installation, some attempts to manage caching fails due to a scalability issue. If you are affected by these issues, manage caching using command line tools as documented in the CLI guides.	X	X	X
Making changes to plugin registration requires a restart of vSphere client. With the Windows client, simply restart the application on your host. The vSphere web client runs as a web service on your VMware vCenter server. Restart the vSphere web client by opening a root ssh window and enter <code>service vsphere-client restart</code> from the command line.	X	X	X
Note that the offload rate bar value might not be accurate.	X	X	X
If the Manage Caching link is not functioning correctly, refresh the page in your browser.	X	X	X
Proper communication between VM in guest caching mode and IO Accelerator Management Tool requires that the VM have a valid IP address assigned to it. Networking must be configured properly on the VM.	X	X	X
When performing the Attach operation, an error message might indicate "Device not found". However, the Attach operation is successfully completed.	X	X	X
Do not detach a cache device. If you must detach the device, deselect it as a caching device, and then detach it.	X	X	X
When formatting a device to factory capacity, the interface indicates that the attach operation is at 10%. The display is incorrect. The attach operation actually does complete, but the task status does not indicate so.	X	X	X
When switching a VM from guest-based caching to no caching, the cache status continues to reflect guest-based caching in the VM grid (at the bottom of the host page) until the VM is rebooted manually.	—	—	X

Issue	Windows	Linux	ESXi
The Caching Status displayed on the Configuration Host page might display a different status than the same cache on the Cache grid.	X	X	—
If you reset stats via CLI, the graphs might show negative numbers for Average IOPS. These values are incorrect.	X	X	—
Guest Based Caching Linux Enabled on VM with Paravirtual SCSI controller causes a failure to assign capacity share to the VM. To resolve this, please replace the virtual ada.	X	X	X
Certain errors in the management logs are expected and normal during host install when the CIM provider is installed but not fully initialized or during a reboot of a host when the connection to the CIM_provider is lost. For more information, see the Support Knowledge Base article.	X	X	—
On the Remote Access page, if user selects a host name that is different from the current hostname, vCenter registration will not be updated. To avoid this, un-register the plug-in, change the host name, and then re-register the plug-in.	X	X	—
VMware might create an issue when attaching or detaching a device that causes system failures. You might be able to resolve this through the CLI, which bypasses the VMware issue.	—	—	X
Anytime you need to restart the <code>fio-msrv</code> process, you might need to run the command twice.	X	X	—
Volumes might be visible in IO Accelerator Management Tool after deleting them from the Windows disk manager.	X	—	—
If you manually remove software from a host, the removal might not be reflected in the UI.	X	X	—
While viewing the Configuration tab, if a Warning or Error link in the Status column of the table is clicked, Navigation proceeds to the Alerts tab. If the Alerts button is clicked, the Alert table is replaced by a Configuration table.	X	X	—
IO Accelerator Management Tool does not load in IE10 when remote access is enabled.	X	—	—
The PCI address might not display correctly in the Info tab.	X	X	—
<code>--list</code> and <code>--directory</code> do not work when used together with <code>fio-remoteupdate.py</code> . To change the directory and then list the available files in the directory, run <code>fioremove-update.py</code> first with <code>--directory</code> , then with <code>--list</code> .	X	X	—

Acronyms and abbreviations

CIM

common information model

FPGA

Field Programmable Gate Array

JSON

JavaScript Object Notation

MLC

multilevel cell (NAND memory type used in SSDs)

MSI

Message Signaled Interrupt

RHEL

Red Hat Enterprise Linux

RPM

Red Hat Package Manager

SLES

SUSE Linux Enterprise Server

VM

Virtual Machine

VMFS

virtual machine file system

VSL

virtual storage layer

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